TKELL KNIVES · DESIGN STUDIO

Installation and user guide

Version 1.4.1 · For Windows and Mac · Send this with the program

> **New in 1.4.1.** A circle’s third click keeps the first tangent. The radius or the center may move so the circle is tangent to both lines. The numbers in a model dialog and in the measure panel sit on a light field. Clicking the tool that is already on, or another sketch or model tool, ends it, the same as Esc. Editing a measured line keeps its constraints, and an offset distance can be typed. The handle and tang open fully on screen, with open space to the right equal to 150% of their length.

> **Drawing tools, from 1.4.0.** Two rows. The first is Handle, File, Build, STEP, DXF, Undo, Redo, Fit, Scan, Chat, and Stock. The second is a Sketch block (Line, Circle, Fillet, Chamfer, Offset, Delete, Move, Break, Trim, Extend) and a Model block (Choil, Ramp, Jimping, Bevel, Swedge, Fillet, Chamfer, Fuller). Nothing is pressed when the knife opens. Click a line to see its length, angle, and constraints, and type or drag them. Sketch tools are click, click, and repeat. Snap stays on unless you hold Ctrl. A Ctrl click never adds a coincident join. Model tools open a dialog. Accepted features collect in a list you can drag.

1. Getting the program

The package is the drawing studio plus the handle and blade master files. It opens in Chrome. It is not a file you double-click.

Download the newest package here. This page is public, so the link can be mailed. It does not ask for a GitHub login:

https://github.com/JPFava/tkell-studio-share/releases/latest

Take both files. This PDF opens by itself. The program is the zip. The zip asks for a password. That password is in the note from TKell. It is not written on the download page.

> **Do not attach the zip to Gmail.** Gmail marks the program “Dangerous” and refuses it, because the folder contains the studio’s code. Send the link above and the password as ordinary text. The person downloads the files themselves.

> **The STEP kernel is not inside the zip.** Drawing, scans, save, and load work as soon as the studio is running. Build, STEP, and DXF also need Build123d, the OpenCASCADE modeler. Install it once on that same Windows or Mac computer. The steps are in Appendix A.

2. Install

What to install first

1. Google Chrome.

2. Node.js 22, from nodejs.org. Take the current 22 release. During the Windows install, leave “Add to PATH” checked.

Windows

1. Unzip `TKell-Design-Studio.zip`. Put the folder somewhere you can keep, such as Documents. Do not run it from inside the zip.
2. Open the folder. Click the address bar, type `cmd`, and press Enter. A command window opens in that folder.
3. Type `npm install` and press Enter. Wait until it finishes. This only has to be done once, and again after a new version.
4. Type `npm run dev` and press Enter. Leave that window open.
5. Open Chrome and go to `http://127.0.0.1:8080`

Mac

1. Unzip the package and move the folder out of Downloads if you want to keep it.
2. Open Terminal. Type `cd ` and then drag the folder onto the Terminal window. Press Enter.
3. Type `npm install` and press Enter.
4. Type `npm run dev` and press Enter. Leave Terminal open.
5. Open Chrome and go to `http://127.0.0.1:8080`

> To stop the studio, click the command window and press Ctrl+C (Windows) or Control+C (Mac). Next time you only need `npm run dev`, then Chrome. You do not run `npm install` again unless you replaced the folder with a newer package.

3. The window

The tools sit on two rows. The first row is the job: handle, file, build, and the view. The second row is Sketch, then Model. Under that is a single view. **2D** is the drawing. **3D** is the solid. The toggle is at the right end of the first row. Only one of them is on screen.

Nothing is pressed when you start. A click on a line measures it. Press a tool when you want to draw or to add a feature. File, Handle, Scan, and Chat close when you pick an item, click Close, or click the drawing.

FIRST ROW	WHAT IT DOES
Handle	The grip the blade is drawn on.
File	Blade name, presets, save, export, import, clear lines, restore.
Build	Writes the STEP and shows it in 3D. Stays dim until the outline is one closed loop.
STEP, DXF	Download the last blank that Build wrote.
Undo, Redo	One action at a time. This is not the same as Restore.
Fit	Zooms the 2D drawing to the knife.
Scan	A photo or drawing under the sketch.
Chat	Type a change in words.
Stock	Blade thickness in inches. Default is 0.165.

SKETCH	WHAT IT DOES
Line, Circle, Fillet, Chamfer, Offset	Click, click, and repeat. Esc, or clicking that button again, or clicking another tool, ends it. Fillet and Chamfer show a distance on the bar. Offset does too.
Delete, Move	Click lines or drag a box, then press Enter. Esc leaves the tool without doing it.
Break, Trim, Extend	One click. Break adds a joint. Trim removes a piece between crossings. Extend runs an open end to the next line.

MODEL	WHAT IT DOES
Choil, Ramp, Jimping, Bevel, Swedge, Fillet, Chamfer, Fuller	Opens a dialog. Click the lines it sits on, set the numbers, then Accept. The item appears in a list you can drag. Click the list to open it again, change it, or delete it. Fillet and Chamfer here are on the solid, not the outline.

The line under the bar is the status. Read it. It tells you what the next click is supposed to do.

4. Start a knife

1. Click **Handle** and pick the grip. The handle and its tang land fully on the 2D board, tip direction to the right. Open space to the right of them is 150% of their length, so there is room for a blade.
2. Click **File**. Set the blade name. That name is used on the saved JSON, the STEP, and the DXF.
3. Still in File, you can start from a preset instead of a blank tang: Drop point, Dagger, Wharncliffe, Tanto, Clip point, or Sheepsfoot. The preset is a sketch on the current handle. You can still edit every line.

The handle you pick is the master grip. Changing handles does not throw away lines you already drew. If the new tang does not meet those lines, use **Move** (below) to shift the drawn blade onto the new tang.

5. Draw the outline

Stay on **2D**. Units on the board are inches. Snap is always on. Hold Ctrl while you click when you do not want the point to stick, and that click will not add a coincident constraint. Esc ends the current tool. Clicking that tool's button again, or another sketch or model button, also ends it. If you had not finished the click, the unfinished piece is dropped. A finished line or circle stays.

Click a line when no tool is on

This is the state you start in. Click a line. A small panel shows the length and the angle. On a curve it also shows the radius, and a dashed chord between the two ends. The angle is the angle of that chord. Any constraint on the line is listed: Coincident, Tangent, or Parallel. Type a new length, angle, or radius, or drag an end to change the length and the angle, and drag the middle of a curve to change the radius. The constraints stay true. A coincident point stays joined. A tangent circle stays tangent, and the radius or the center may move to keep it. A parallel offset stays parallel. If the line is an offset, the panel also shows that distance so you can type a new one. Esc clears the selection.

Line

The first click starts the line. If that click is an endpoint of something already drawn, the new line is coincident there. The next click creates the line. If that click is also an endpoint, the line is coincident and Line stops. If it is not, another line starts from that point and keeps going until you click an endpoint or press Esc. Esc before the second click cancels. Esc after a finished segment keeps the line and leaves the tool.

Hold Alt (Option on a Mac) and drag a point when you want to move it instead of drawing. Hold Alt on a gold dot to move that end of the tang. A sketch point already sitting on that end moves with it. Grab the middle of a line and drag to bow it into a curve. A radius shorter than half the distance between the two ends is refused.

Circle

The first click is the center. If that click is an endpoint, the center is coincident. The next click sets the radius to the distance between the two clicks. If that click lands on a line or a curve, the circle is tangent to it. When the center was free and the radius click was tangent, a third click on another line keeps that first tangent and adds the second. The radius or the center may move so both tangents stay true. If the two lines cannot share a circle, the circle stays as it is. Esc instead of the third click also keeps it. A coincident center, or a free center whose radius click is not tangent, finishes the circle in two clicks. Esc before the second click cancels. Esc after the circle exists keeps it. Drag the middle of an arc later to change the radius, or click the arc with no tool on and type it.

Fillet

This rounds a corner of the drawing. It changes the outline. Click two lines or curves. They cannot be parallel. Each one is extended from the nearer end until the two meet. A fillet of the radius on the bar, default 0.060 in, is placed at the two tangent points nearest that corner, and the pieces between the tangent points and the corner are trimmed away. Click another pair, or press Esc. Esc before the second line abandons the first.

Chamfer

This is the sketch chamfer, on the outline, not the solid. Click two lines or curves. They cannot be parallel. They extend until they meet. A straight flat is cut, set back along both lines by the distance on the bar, default 0.060 in. The status reports the length of that flat. Click another pair, or press Esc.

Offset

Click the line or curve, then click the side to copy toward. A parallel copy is placed at the distance on the bar, default 0.060 in, and a parallel constraint is stored. Click the line later, with no tool on, to type a different offset. Click another line, or press Esc.

Delete

Click Delete, or press the Delete key. That only starts the tool. It does not delete yet. Click lines, or drag a box around them. Press Enter to delete them. Constraints that belonged to those lines are removed. Esc leaves Delete without deleting.

Move

Click Move. Click lines or drag a box until the selection is right, then press Enter. Click the from point, then the to point. The selected lines move by that shift. A constraint between a line you moved and a line you left behind is dropped, and a shared corner is split so the line you left stays put. Constraints among the lines you moved stay. Esc leaves Move.

Break, Trim, and Extend

Break: click a line or curve where you want a joint. One line becomes two, and they share that point. Trim: click the piece you want gone. It stops at the crossings. Extend: click an open end. It runs straight until it meets another line.

When Build will light up

Build stays unavailable while the outline is open. The button shows how many open loops are left. You need one closed outline. A closed loop drawn inside that outline is a cutout. An open loop that starts and ends on the outline and bulges outside it is an addition, such as a ring or a glass breaker. The glass breaker on the Agent tang is part of the handle master; do not trim it off when you only meant to change the blade.

Undo and **Redo** step one action. **Restore**, under File, puts the knife back to the starting sketch. It is not an undo.

6. Features

Click a Model tool. A dialog opens. Click the drawing to choose the lines, and click a listed line in the dialog to drop it. Set the numbers, then Accept. Cancel closes the dialog and does not add the feature. Every accepted item is added to a small list titled Model. Drag that list by its title. Click an item to open its dialog again, change the numbers, or delete it.

Choil and thumb ramp are cut out of the blade. They are not material added on the outside. Sketch Fillet and sketch Chamfer change the outline. Model Fillet and Model Chamfer do not. They break the corner of the solid.

TOOL	HOW YOU PLACE IT	WHAT YOU SET
Choil	One click on a line.	Radius. An X marker stays on the line and can be dragged.
Ramp	One click on a line.	Radius and height of the thumb ramp.
Jimping	Two clicks, the start and the end of the run.	Radius, how far the grooves sit below the line, and the count. Each groove has sides tangent to the radius, so a flat shows between scallops. Two X markers mark the ends.
Bevel	Two clicks along the edge. The run can cross several connected segments. The tip is usually inside the bevel.	See the bevel section below.
Swedge	Two clicks along the spine.	Height at one end, zero at the other. The tall end can be the start or the stop. If a bevel is in the way, the swedge stops where they meet. This is a triangle, not a flat grind.
Chamfer	Two clicks along a corner.	Angle, default 45°, plus a height at the start and a height at the stop. 45° takes the same amount off the face and into the thickness. This flattens the corner. It does not change the outline.
Edge fillet	Two clicks, the start and the stop, the same way as a chamfer. The run can cross several connected segments. In the Model block this tool is labeled Fillet.	Radius at the start and radius at the stop. Default is 0.030 in at both ends. This is a quarter-round: the radius is on the face and the same distance into the thickness. The two ends can differ. It does not change the outline. Accept, then Build to see the round in the solid.
Fuller	Draw a closed loop inside the outline, then click Fuller and that loop.	Depth and the radius in the bottom. Depth 0 makes it a through cutout again.

Bevel

Defaults are a 20° primary, a 40° secondary, an edge flat of 0.018 in, and a secondary meet of 0.040 in. Both angles are **inclusive**. A 20° primary is 10° each side of center, not 20° each side.

- **Edge flat** is the thickness left at the cutting edge. The primary starts on center, set back by half of that flat, on both sides.
- **Sharpen** is on by default. Turn it off for a flat grind with no secondary bevel. A swedge is a separate tool. Do not fake a swedge by turning Sharpen off.
- **Secondary by angle** runs the secondary out from center until it meets the primary. The panel tells you the height where they meet.

- **Secondary by height** makes them meet at the distance you type, and the panel tells you the angle that takes.
- The bevel height stays parallel to the edge. It does not shrink just because the tip is near. If the blade runs out, the grind runs out with it.
- **Plunge radius** is a ball-end mill held square to the face. The radius is constant. The depth goes from full at the bevel to zero where the grind meets the flat. Plunge angle is from square: −5° to +10°. Zero is perpendicular. Positive leans toward the tip.
- Where two bevels continue into each other, there is no plunge between them. The plunge is only at the end that is not the tip.

The 2D view shows the bevel landing at full thickness. The 3D view, after Build, is the grind in the STEP, not a picture painted on the blank.

Chamfer and edge fillet

Both sit on a corner of the outline. Neither one moves the line you drew. A chamfer takes a flat off that corner. An edge fillet puts a quarter-round there instead. Set the angle and the two heights on a chamfer. Set the two radii on an edge fillet. Neither one has a plunge. In the drawing, a chamfer is a green dashed mark and an edge fillet is a lilac dashed mark. The solid shows them only after Build.

7. Build, STEP, and DXF

1. Close the outline. Build is clickable when it no longer shows an open-loop count.
2. Set **Stock** if this blade is not 0.165 in.
3. Click **Build**. The view switches to 3D. The status counts the seconds. Leave it alone until it finishes. Clicking Build again starts the job over.
4. The 3D view is orthographic. Drag to orbit. The lamp can be dragged so a swedge or a fuller is easier to see. Click **2D** to go back to the drawing. The solid stays.
5. **STEP** downloads the blade blank. **DXF** downloads the laser profile. Both use the blade name.

> If Build says Build123d is not installed, finish Appendix A, then click Build again.

8. A scan or a sketch under the knife

Click **Scan**, then **Load scan**. Photos, PNG, JPG, WEBP, and iPhone HEIC files are accepted.

1. Click **Match**. On the picture, click a point or drag a box around a hole. Do the same for a second hole. Then pick the same two holes on the handle model, in the same order. With Snap on, the center of a handle hole will catch the click.
2. Which end you pick first does not have to be a particular hole, but the second pair must be the same two holes in the same order.
3. If the artwork is left-handed and the studio handle is the usual right-handed view (handle left, tip right), turn **Mirror** on before you match, or the drawing will land upside down.
4. A box around a hole looks for one circle. If it fails, the status says why: not enough contrast, not round enough, or more than one circle. A few stray line bits are tolerated. A second full circle in the

box is not.

Hand drawings will not scale perfectly. After the match, nudge about the first handle point. Each control is independent, and you can go back and forth:

- Rotate in steps of 0.01°, limited to ±2.00°.
- Shift one picture pixel at a time. Hold the button to keep moving.
- Scale from 0.900× to 1.100× in steps of 0.001.

Undo nudge and Redo nudge only undo those nudges. They do not undo the knife. Holes help, but the place the drawn blade meets the handle matters more. Nudge until that junction is right.

Construction geometry

After the scan is matched and Show scan is on, click **Add construction geometry** and drag a box around ink you want to trace. Lines, curves, and endpoints in that box become snap points. They are not part of the knife. Show construction turns them on and off. They move with later nudges. Use them to draw a cutout or an extension that follows the artwork. This command is only there after a scan is placed.

9. Save, export, and send a design to someone else

Under **File**:

- **Save** keeps the blade name, the handle, and the sketch on this computer.
- **Export** writes a JSON file. It includes the sketch, the scan picture, the hole match, the nudges, and the construction geometry. The file name includes the blade name. This is the file to email another designer.
- **Import** opens that JSON. The 2D drawing comes back, including the scan.

Clear lines deletes the sketch and leaves the handle. **Restore** returns to the original sketch for this session. Neither one is Undo.

10. Chat

Open **Chat** and type the change the way you would say it at the bench. Examples: “put a dagger on the Agent grip”, “remove the guards”, “lengthen the blade half an inch”, “narrow the belly”. Send adds your line and the studio’s reply to the same conversation, so you can keep refining. **+ New chat** starts over. The chat does not replace the drawing tools. If a result is wrong, Undo is still one click on the bar.

A dagger on a grip means the heel through the ricasso stays the grip’s own tang, and the blade forward of the ricasso is the dagger, blended at the ricasso. The grip scales stay the grip you had. Guards on a dagger blank are quillons. “Remove the guards” clips those off. It does not cut the glass breaker off the Agent tang. Everything behind the ricasso stays.

11. A short first session

1. Handle → the grip you are designing for.
2. File → name the blade. Optionally pick a preset.

3. Line → close an outline from the tang, around the tip, and back. Click an endpoint to finish the chain, or press Esc to keep what you have drawn. Use Circle for a true radius. Use Fillet on a square corner.
4. Bevel → the dialog opens. Two clicks along the edge, then Accept. Leave the defaults unless the work order says otherwise.
5. Swedge → two clicks on the spine if the tip needs one, then Accept. Set which end is tall.
6. Chamfer or Fillet in the Model block → two clicks on a corner that should be broken flat or rounded. Set the numbers, then Accept. These do not move the outline.
7. Build → wait. Orbit the 3D view. Drag the lamp if the grind is hard to see.
8. File → Save, then Export if someone else needs the design.

12. When something is wrong

WHAT YOU SEE	WHAT TO DO
Build stays dim and shows a number of open loops.	The outline is not closed, or an extra chain is still open. Join the free ends. One closed loop is required.
A choil or ramp sits outside the steel.	Those features cut material away. Place them on the outline, not out in space. Snap is already on. Hold Ctrl only when you mean to miss the point.
The scan is upside down.	Mirror, then match the two holes again in the same order.
The scan is close but the junction is off.	Nudge. Rotate, shift, and scale are separate. Undo nudge if you overshoot.
Build says it is still working.	Wait. The status counts seconds. Do not click Build again unless you intend to start over.
STEP or DXF downloads an older blade.	Build again after the last change, then download.
Chrome will not open the page.	The command window must still be running `npm run dev`. The address is `http://127.0.0.1:8080`
A click keeps jumping to the wrong point.	Hold Ctrl for that click. Snap stays off only while Ctrl is down, and that click does not add a coincident constraint.
A click on an endpoint starts a line when you meant to drag it.	Leave Line first, with Esc. With no tool on, drag the end. Or hold Alt (Option on a Mac) while Line is on, and keep holding it while you drag.
Fillet or Chamfer says the lines are parallel.	Parallel lines never meet. Pick two that can cross if you extend them.
Delete removed nothing.	Delete only starts the tool. Click the lines, or drag a box, then press Enter.
The tang end needs to move.	Hold Alt and drag the gold dot at that end.
Trim says the line does not cross another line.	Trim only removes a piece that meets another line. Extend an open end, or delete the line.
Fillet says the radius is longer than the line, or the corner is too flat.	Type a smaller radius in the Fillet field on the bar, then click the two lines again.
An edge fillet does not show on the solid.	It does not change the outline. Click Build after you place it. The lilac mark on the drawing is only the location.
Build says Build123d is not installed.	The STEP kernel is missing. Follow Appendix A, then click Build again.

WHAT YOU SEE	WHAT TO DO
A new package was just unzipped over the old folder.	Run `npm install` once, then `npm run dev`. Leave the `.cad` folder in place if it is already there.

TKell Design Studio 1.4.1. This guide matches that version. Newer packages stay at the latest-release link above.

Appendix A. Install the STEP modeler on Windows or Mac

Build writes a real STEP solid. The modeler is Build123d, which sits on OpenCASCADE. It is the same kernel the shop Linux machine uses. It is not in the zip because the download is large and it has to match the computer it runs on. You install it once, inside the studio folder, in a folder named `.cad`. The studio looks there automatically. On Mac that file is `.cad/bin/python`. On Windows it is `.cad\Scripts\python.exe`.

Use Python 3.12 from python.org. Build123d's geometry library publishes installers for Python 3.11, 3.12, 3.13, and 3.14, on Windows 64-bit and on both Apple Silicon and Intel Macs. Python 3.12 is the one these steps use. Do not use the Python from the Microsoft Store, and do not use the old Python that comes with macOS.

You need an internet connection the first time. The download is a few hundred megabytes. Leave about 2 GB free. The first Build after installing can take a couple of minutes. Later builds are the same speed as on the shop machine.

A.1 Windows

1. Download the Python 3.12 Windows installer (64-bit) from python.org/downloads/windows. Run it. On the first screen, check **Add python.exe to PATH**, then choose Install Now.
2. Unzip the studio if you have not already, and open a command window in that folder. Click the folder's address bar, type `cmd`, and press Enter.
3. Confirm Python answers, then create the modeler beside the studio. Paste this whole block:

```
py -3.12 -m venv .cad
.cad\Scripts\python.exe -m pip install --upgrade pip
.cad\Scripts\python.exe -m pip install build123d pyyaml
```

4. Wait until pip says it successfully installed build123d. A red error about a missing Microsoft C++ compiler means pip tried to compile instead of downloading a ready-made wheel. That usually means the Python is not 3.12 64-bit. Remove the `.cad` folder and repeat from step 1 with the 64-bit 3.12 installer.
5. Check the install:

```
.cad\Scripts\python.exe -c "import build123d, yaml; print('ready')"
```

It should print `ready`. Then start the studio with `npm run dev` if it is not already running, open Chrome at `http://127.0.0.1:8080`, and click Build.

> Windows hides folders that start with a dot, so you may not see `.cad` in File Explorer. That is fine. Do not delete it when you update the studio. If you replace the whole folder with a new zip, run the three commands above again.

A.2 Mac

1. Download the Python 3.12 macOS installer from python.org/downloads/macos. Use the universal installer. It covers Apple Silicon and Intel. Open the pkg and install it. If macOS blocks it, open System Settings → Privacy & Security and allow the installer.

2. Open Terminal. Type `cd ` (with the space) and drag the studio folder onto the window. Press Enter.

3. Create the modeler and install it:

```
python3.12 -m venv .cad
.cad/bin/python -m pip install --upgrade pip
.cad/bin/python -m pip install build123d pyyaml
```

4. Check it:

```
.cad/bin/python -c "import build123d, yaml; print('ready')"
```

It should print `ready`. Then `npm run dev`, Chrome at `http://127.0.0.1:8080`, and Build.

If Terminal says `python3.12: command not found`, the installer did not finish or Terminal was opened before the install. Close Terminal, open it again, and retry. The python.org install also places a copy at `/Library/Frameworks/Python.framework/Versions/3.12/bin/python3`. You can use that full path in place of `python3.12` on the first command only.

A.3 If the install fails

WHAT YOU SEE	WHAT TO DO
`py` or `python3.12` is not recognized.	Python 3.12 is not installed, or the command window was opened before the install. Install it from python.org, close the window, and start again in the studio folder.
pip talks about building a wheel, or asks for a C++ compiler.	The Python is the wrong version or not 64-bit. Delete the `.cad` folder and repeat with Python 3.12 64-bit.
A download stops or times out.	The OpenCASCADE wheel is large. Run the last pip command again. It resumes.
The check does not print `ready`.	Read the last lines of the error. If it cannot import build123d, the pip install did not finish. Run it again.
Build still says Build123d is not installed.	The `.cad` folder is not inside the studio folder, or the studio was started from a different folder. Stop it with Ctrl+C, `cd` to the studio folder, and run `npm run dev` again.

You do not install a separate Linux program, WSL, or a virtual machine. The Windows and Mac packages of this same modeler are what Build calls.